

3(a) The electric field strength at a point is defined as the electric force exerted per unit positive charge placed at that point.

(b)(i) (Before $x = 10$ cm, the electric field points to the right. After 10 cm, it points to the left.) Since the electric field changes direction between A and B, the charges have the same sign.

OR

At a point between A and B, the electric fields due to each charge cancel out to give a resultant field strength of zero.

So the charges have the same sign.

(ii) Any 1 point:

- Charge A is not the only charge present.
- The electric field strength is also influenced by charge B.
- The electric field strength is due to two/both charges.
- The electric field strength is the resultant of the two fields due to charges A and B.

(iii) From the graph, $E = 1.8 \times 10^3 \text{ N C}^{-1}$ where $x = 6.0$ cm.

Electric force on proton, $F_E = qE = (1.60 \times 10^{-19})(1.8 \times 10^3) = 2.88 \times 10^{-16} \text{ N}$

Acceleration of proton, $a = \frac{F_E}{m} = \frac{2.88 \times 10^{-16}}{1.67 \times 10^{-27}} = 1.72 \times 10^{11} \text{ m s}^{-2}$

(iv) At $x = 10$ cm, field strength due to A = field strength due to B (in magnitude)

$|E_A| = |E_B|$ [CONCEPT]

$$\frac{Q_A}{4\pi\epsilon_0 r_A^2} = \frac{Q_B}{4\pi\epsilon_0 r_B^2}$$

$$\frac{Q_A}{Q_B} = \frac{r_A^2}{r_B^2} = \frac{(10.0 \text{ cm})^2}{(5.0 \text{ cm})^2} = 4$$

4(a) Magnetic force $F_m = Bev = (4.0 \times 10^{-3}) \times (1.60 \times 10^{-19}) \times (0.60 \times 10^3) = 3.84 \times 10^{-25} \text{ N}$